SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.

- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway. • Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64 • Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64 • Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structure d code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Question 1

Given

- IP Address = 192.168.10.65
- Subnet Mask = 255.255.255.192 (/26)
- Default Gateway = 192.168.10.1

Solution

A /26 subnet has a block size of 64.

The subnet ranges are:

- 192.168.10.0 – 192.168.10.63

- 192.168.10.64 – 192.168.10.127
- 192.168.10.128 – 192.168.10.191
- 192.168.10.192 – 192.168.10.255

The configured IP address 192.168.10.65 belongs to the subnet 192.168.10.64/26.

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Valid Host Range: 192.168.10.65 – 192.168.10.126

The configured default gateway (192.168.10.1) belongs to the subnet 192.168.10.0/26, which is different from the workstation's subnet. Therefore, the gateway configuration is incorrect.

Configuration Error

The IP address is valid, but the default gateway is in a different subnet. Because of this, the workstation cannot communicate properly with other devices.

Optimized Configuration

IP Address = 192.168.10.66

Subnet Mask = 255.255.255.192

Default Gateway = 192.168.10.65 *(or any valid router interface within the same subnet, such as 192.168.10.126 if configured)*

Usable Host Addresses

$= 2^{(32-26)} - 2$

$= 64 - 2$

= 62 hosts

Explanation

The problem occurs because the default gateway must always belong to the same subnet as the host. After correcting the gateway, the workstation can communicate with other devices in the network successfully.

Question 2

Given

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Solution

Expanded IPv6 Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

With a /64 prefix:

Device A

2001:0db8:0000:0000::/64

Device B

2001:0db8:0000:0000::/64

Both devices have the same /64 network prefix, so they belong to the same subnet.

However, if communication still fails, the issue is likely due to incorrect interface configuration, routing, firewall settings, or Neighbor Discovery Protocol rather than the IPv6 addresses themselves.

Optimized Configuration

Device A = 2001:db8::10/64

Device B = 2001:db8::20/64

Both devices remain in the same subnet with simpler addresses.

Available Host Addresses

Host bits = 64

Number of host addresses

= 2^{64}

= 18,446,744,073,709,551,616

Explanation

Both devices already belong to the same IPv6 network because their first 64 bits are identical. Assigning simple IPv6 addresses makes network management easier while maintaining proper communication within the subnet.

Question 3

Given

Network = 192.168.1.0/24

Subnets used:

- /26
- /27
- /28

Solution

Subnet 1 (/26)

Network Address = 192.168.1.0

Host Range = 192.168.1.1 – 192.168.1.62

Broadcast Address = 192.168.1.63

Usable Hosts = 62

Subnet 2 (/27)

Network Address = 192.168.1.64

Host Range = 192.168.1.65 – 192.168.1.94

Broadcast Address = 192.168.1.95

Usable Hosts = 30

Subnet 3 (/28)

Network Address = 192.168.1.96

Host Range = 192.168.1.97 – 192.168.1.110

Broadcast Address = 192.168.1.111

Usable Hosts = 14

Analysis

Some departments have more IP addresses than they actually need, while others do not have enough. This results in poor utilization of available IP addresses.

Debugging

The subnet allocation is inefficient because fixed subnet sizes are being used without considering the actual number of devices in each department.

Optimized Solution

The best solution is to implement VLSM (Variable Length Subnet Masking). VLSM assigns larger subnets to departments with many devices and smaller subnets to departments with fewer devices. This improves IP address utilization and leaves more addresses available for future expansion.

Explanation

Using VLSM helps reduce IP address wastage and makes the network more flexible. It allows each department to receive only the number of IP addresses it actually requires, improving overall network efficiency.

Question 4

Given

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Solution

Expanded IPv6 Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

With a /64 prefix, the first 64 bits represent the network.

Device A Network Prefix

2001:0db8:0000:0000::/64

Device B Network Prefix

2001:0db8:0000:0000::/64

Both devices belong to the same IPv6 network because their first four blocks are identical.

Debugging

Since both devices are in the same subnet, the communication problem is not caused by the IPv6 addresses. The issue may be due to an incorrect interface configuration, disabled IPv6 settings, firewall restrictions, or problems with the Neighbor Discovery Protocol (NDP).

Optimized Configuration

A simpler addressing scheme can be used.

- Device A → 2001:db8::10/64
- Device B → 2001:db8::20/64

These addresses belong to the same subnet and are easier to manage.

Available Host Addresses

Host bits = 64

Number of available host addresses

$= 2^{64}$

= 18,446,744,073,709,551,616

Explanation

The IPv6 addresses are already part of the same network, so the addressing is not the actual problem. Checking interface status, routing configuration, and Neighbor Discovery settings is the correct debugging approach. Using simple IPv6 addresses also makes network administration easier.

Question 5

Given

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP = 192.168.2.100

Solution

The destination IP address is 192.168.2.100.

Using the longest-prefix match, the router compares the destination address with all entries in the routing table.

The destination belongs to the network:

192.168.2.0/24

Therefore, the matching routing table entry is:

192.168.2.0/24

The packet will be forwarded to the next-hop router:

10.0.0.3

The destination IP is valid because it falls within the host range of the 192.168.2.0/24 network.

Host Range:

192.168.2.1 – 192.168.2.254

Broadcast Address:

192.168.2.255

Debugging

Since the routing entry already exists, the routing table is correctly configured. If users still cannot reach the destination network, the problem could be caused by:

- Incorrect next-hop configuration
- Interface failure

- Missing return route
- Firewall or ACL blocking the packets

Optimized Routing

The router should verify that the next-hop address 10.0.0.3 is reachable and that the routing table is updated correctly. Dynamic routing protocols such as OSPF or EIGRP can also be used to automatically update routing information and improve network reliability.

Default Route

If no matching route is found, the router forwards the packet to the default gateway 10.0.0.1.

Explanation

Routers always use the longest-prefix match to select the most specific route. Since the destination belongs to 192.168.2.0/24, the router forwards the packet through 10.0.0.3. The default route is used only when no specific route is available. This ensures efficient packet forwarding and reliable communication across different networks.